

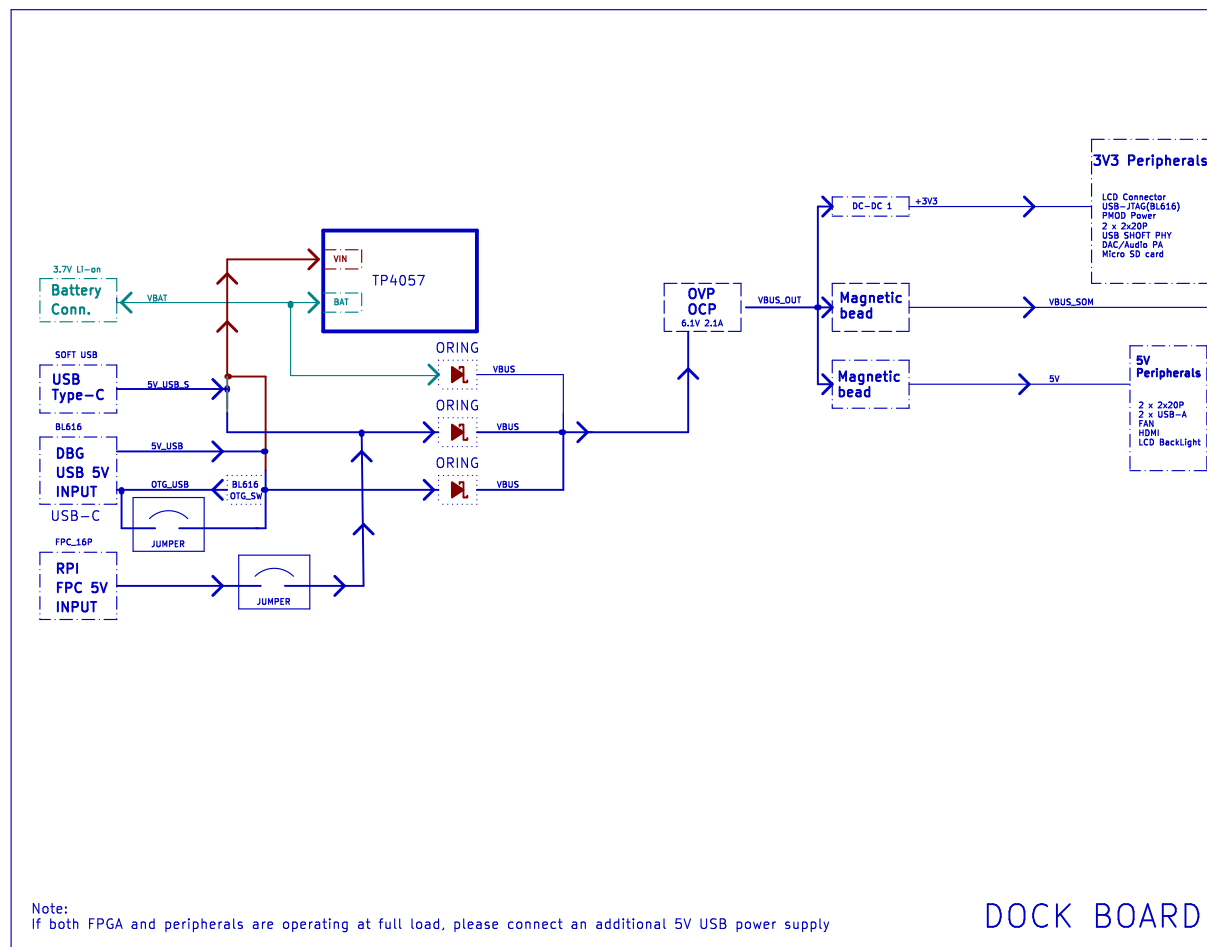


**Shenzhen Sipeeed Technology Co.,Ltd.**

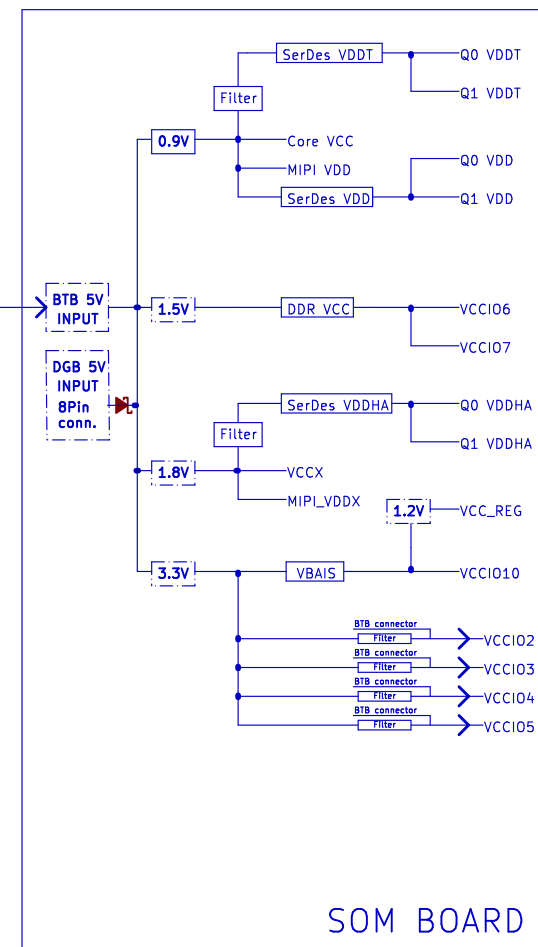
**Title:** Tang\_60K\_Console\_V32001A

Rev: 1.22

Id: 1/17



DOCK BOARD



SOM BOARD



A Retro Ready Console Powered by FPGA

Shenzhen Sipeed Technology Co., Ltd.

Sheet: /POWER\_TREE/

File: PWR\_TREE.kicad\_sch

Title: Tang\_60K\_Console\_V32001A

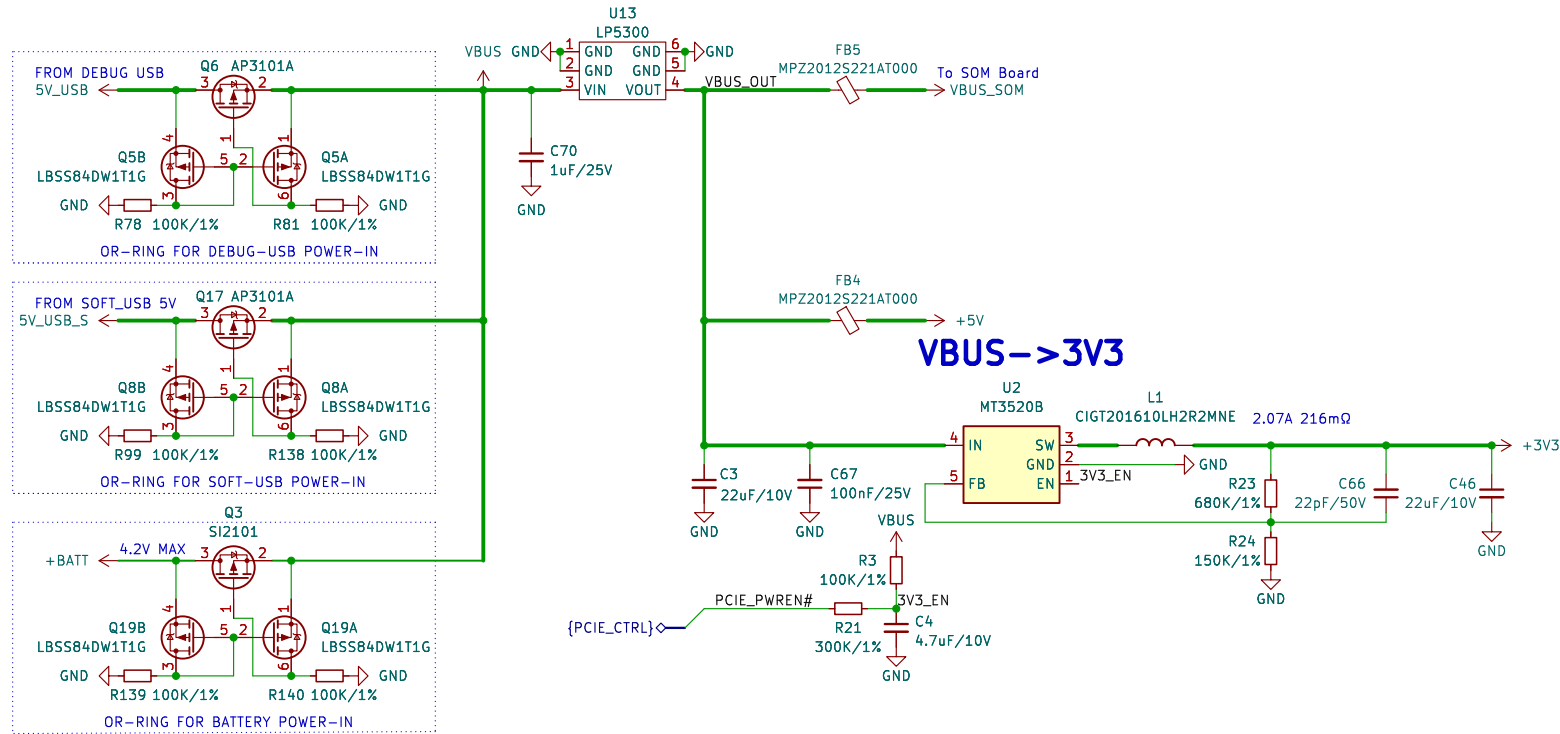
Size: A4 Date: 2024-12-31

KiCad E.D.A. 9.0.1

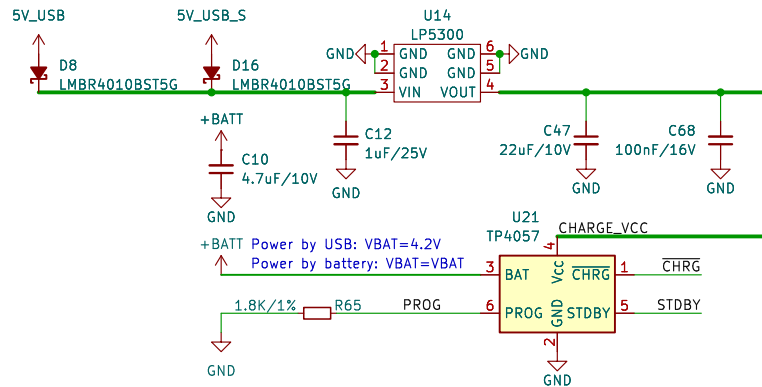
Rev: 1.22

Id: 2/17

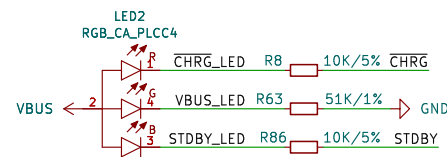
# SYS POWER



## Battery CHRG



## VBUS\_LED



Connect to  
3.7V Li-on battery



A Retro Ready Console Powered by FPGA

Shenzhen Sipeed Technology Co., Ltd.

Sheet: /POWER/

File: PWR\_MAIN.kicad\_sch

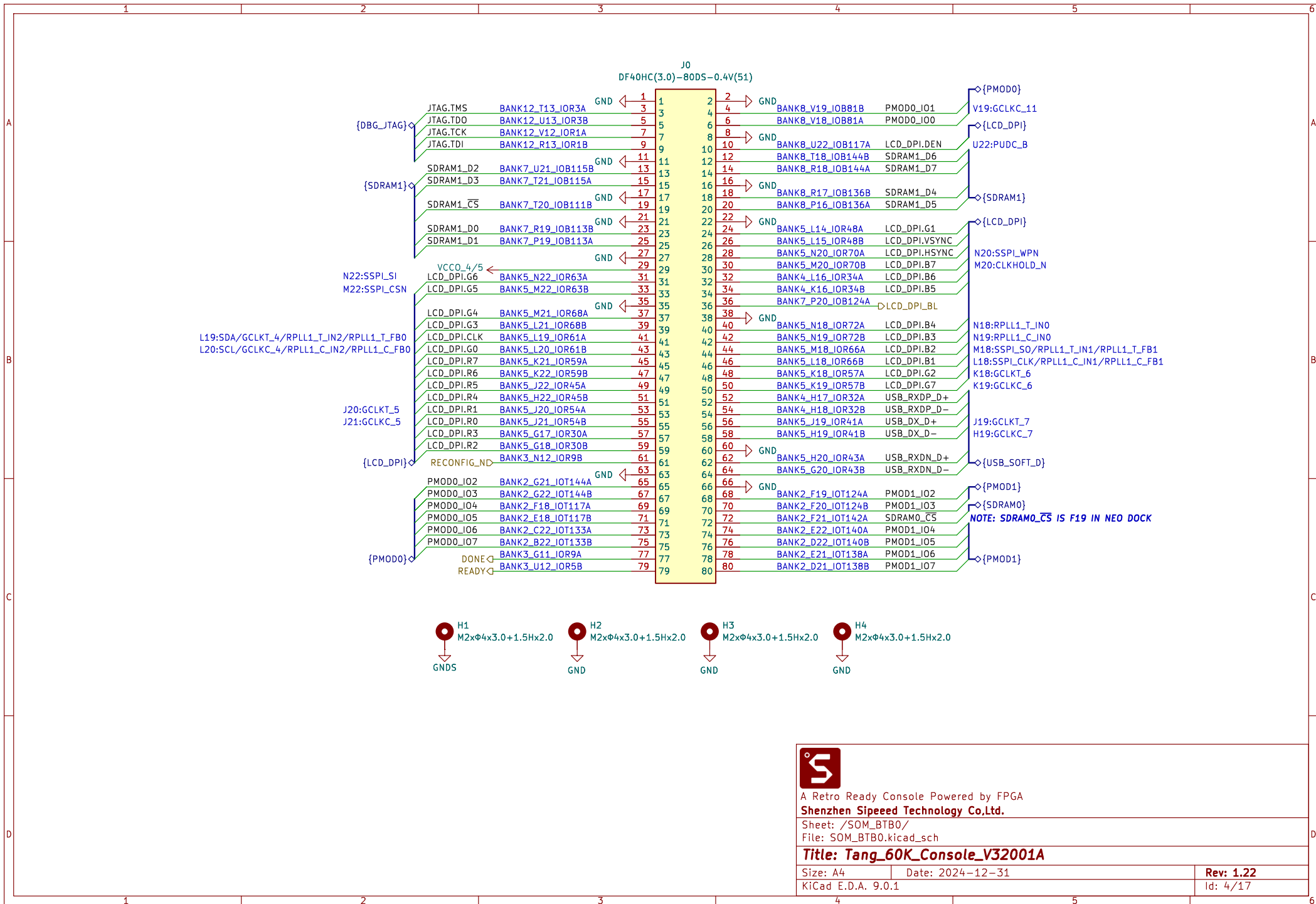
Title: Tang\_60K\_Console\_V32001A

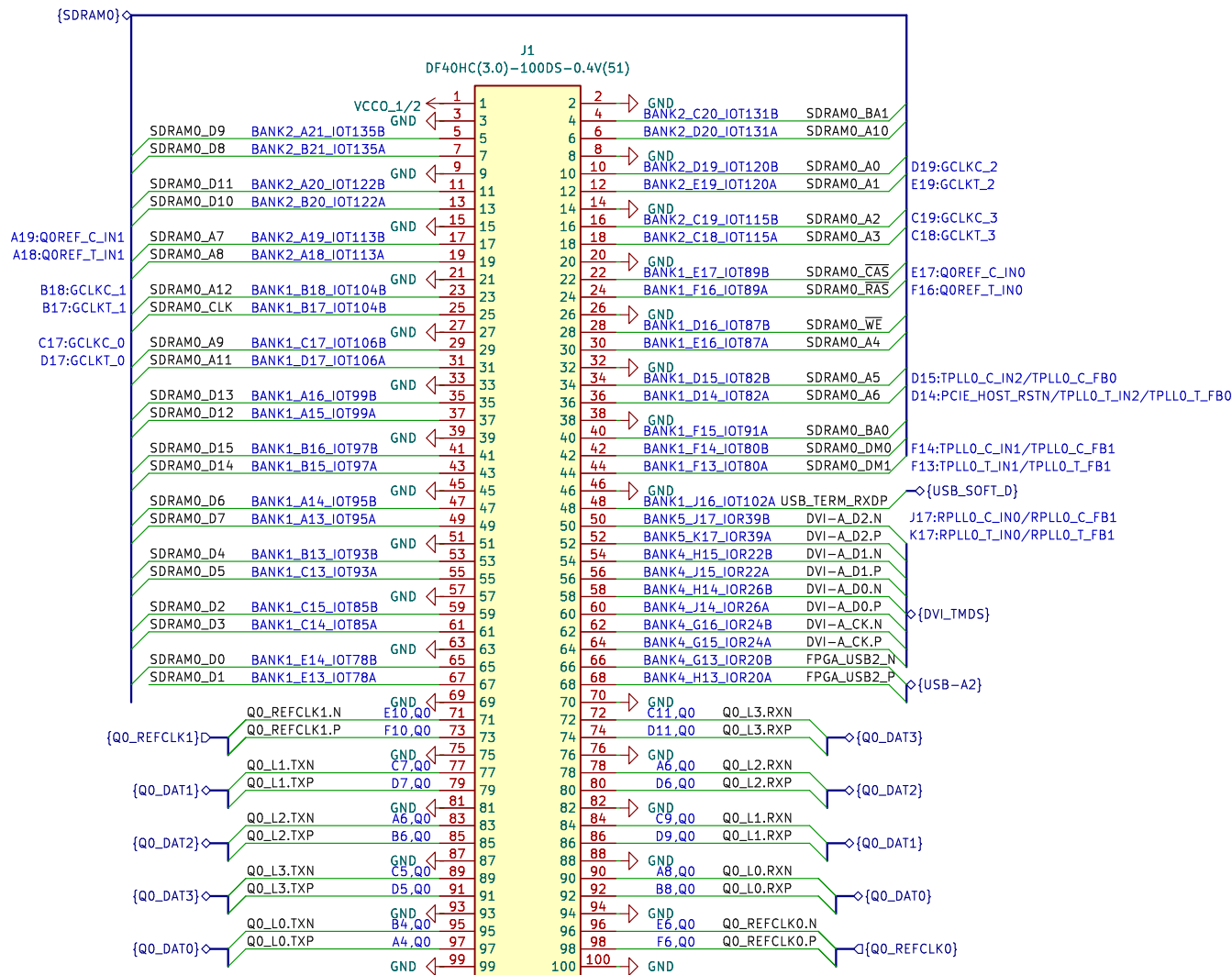
Size: A4 Date: 2024-12-31

KiCad E.D.A. 9.0.1

Rev: 1.22

Id: 3/17





A Retro Ready Console Powered by FPGA

Shenzhen Sipeed Technology Co.,Ltd.

Sheet: /SOM\_BTBI/

File: SOM\_BTBI.kicad\_sch

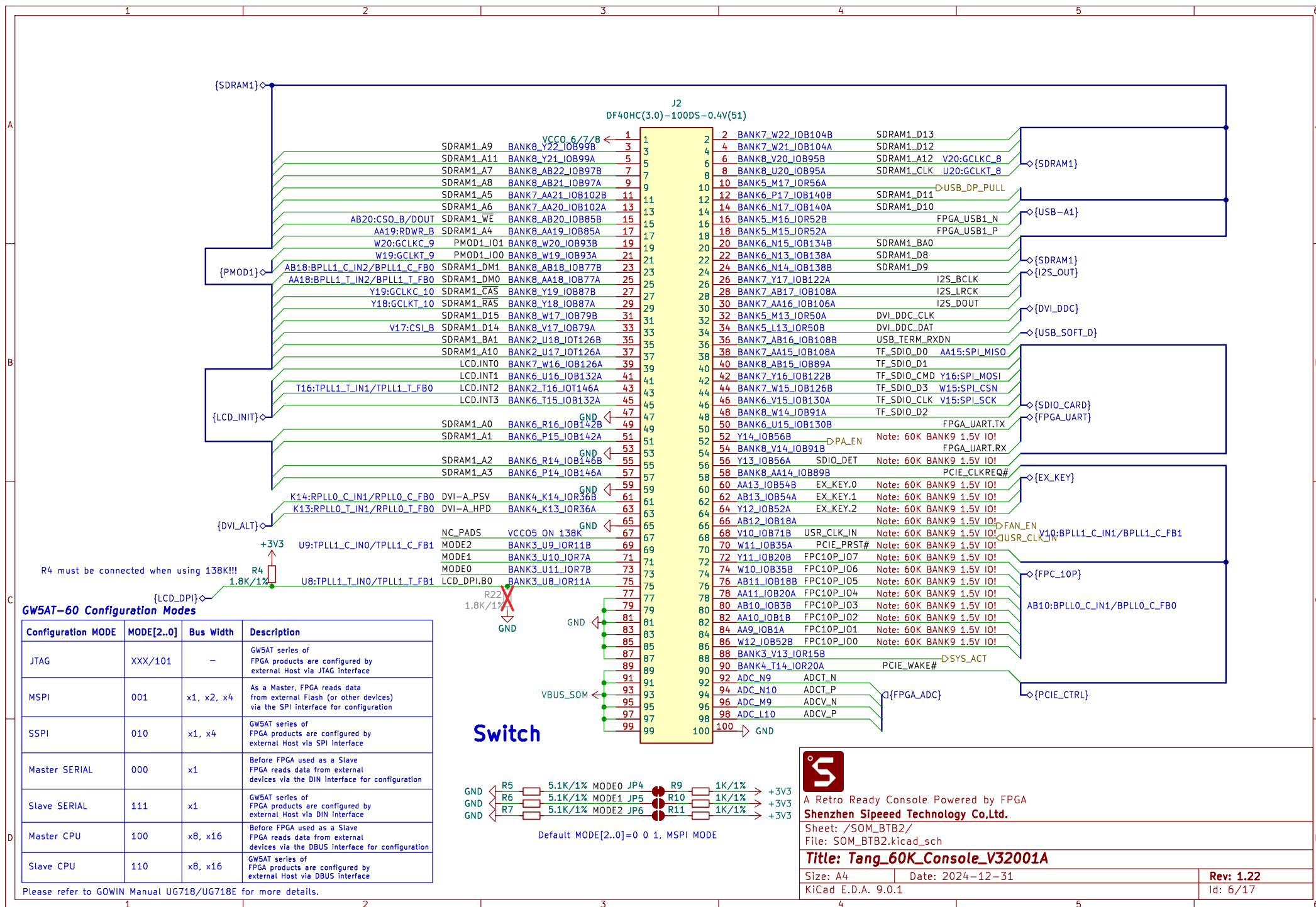
Title: Tang\_60K\_Console\_V32001A

Size: A4 Date: 2024-12-31

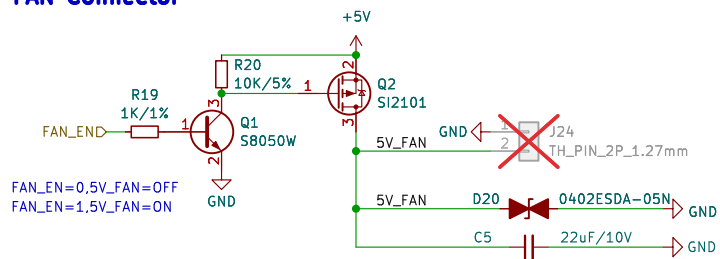
KiCad E.D.A. 9.0.1

Rev: 1.22

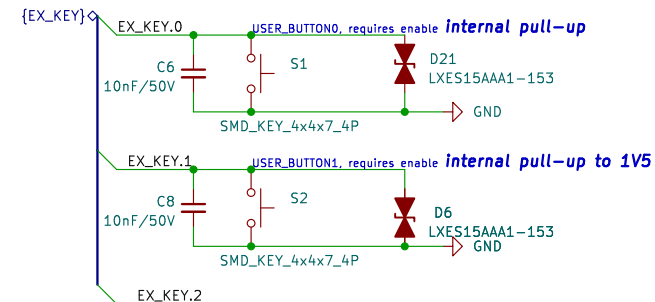
Id: 5/17



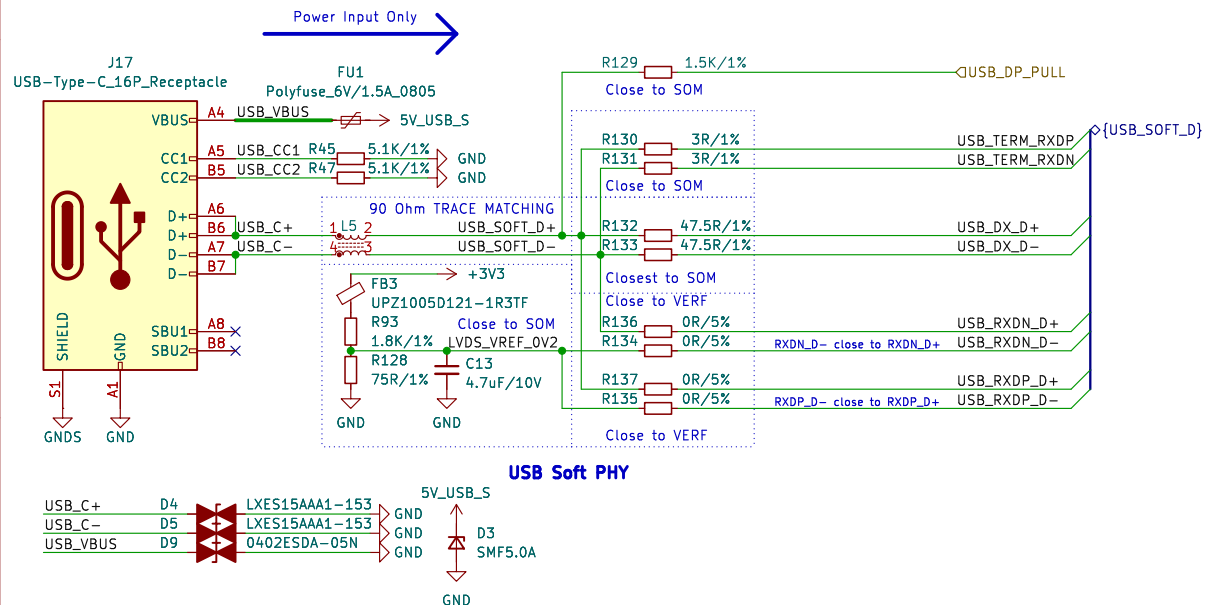
## FAN Connector



**BUTTON**



## USB Soft Deivce



## A Retro Ready Console Powered by FPGA

Shenzhen Sipeeed Technology Co.,Ltd.

Sheet: /PERIPHERALS/

File: PERIPHERALS.kicad\_sch

**Title:** Tang\_60K\_Console\_V32001A

|          |                  |
|----------|------------------|
| Size: A4 | Date: 2024-12-31 |
|----------|------------------|

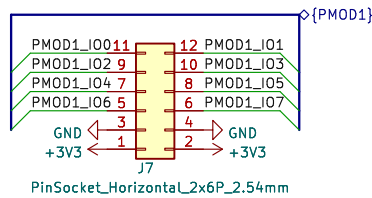
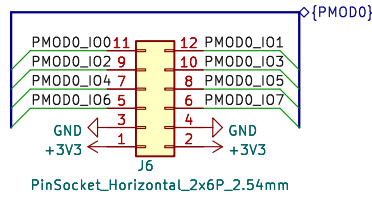
KiCad E.D.A. 9.0.1

Rev: 1.22

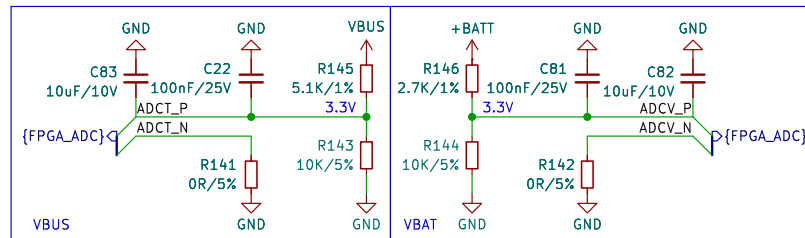
Id: 7/17

## EX CONN.

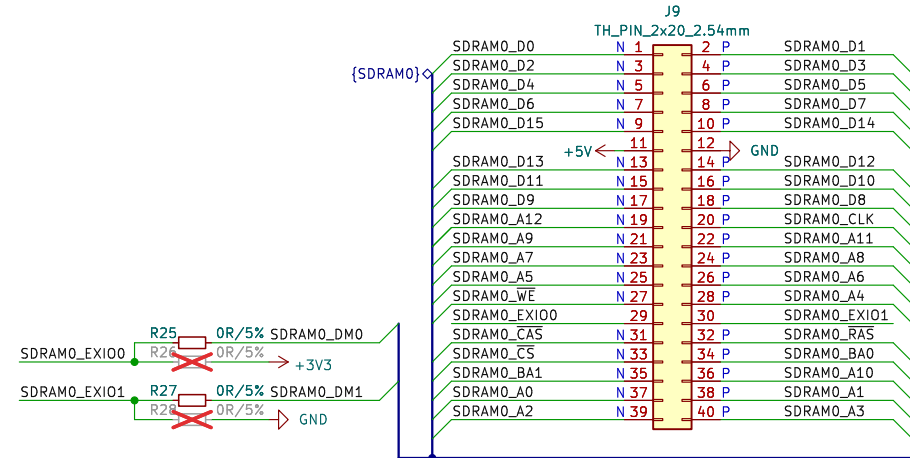
### PMOD



### ADC INPUT



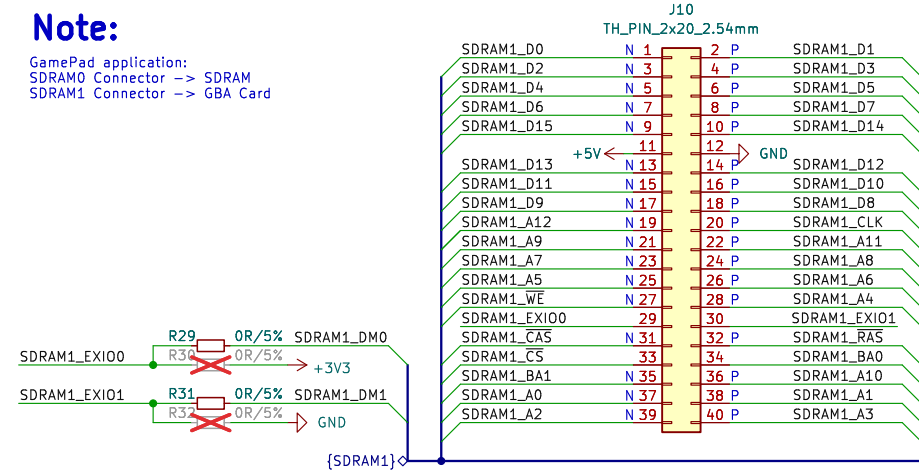
### SDRAM0 CONN. Close to PCIE connector



### Note:

GamePad application:  
SDRAM0 Connector -> SDRAM  
SDRAM1 Connector -> GBA Card

### SDRAM1 CONN. Close to PMOD connector



A Retro Ready Console Powered by FPGA

Shenzhen Sipeed Technology Co., Ltd.

Sheet: /FPGA\_EXT\_CONN/

File: EX\_CONN.kicad\_sch

Title: Tang\_60K\_Console\_V32001A

Size: A4 Date: 2024-12-31

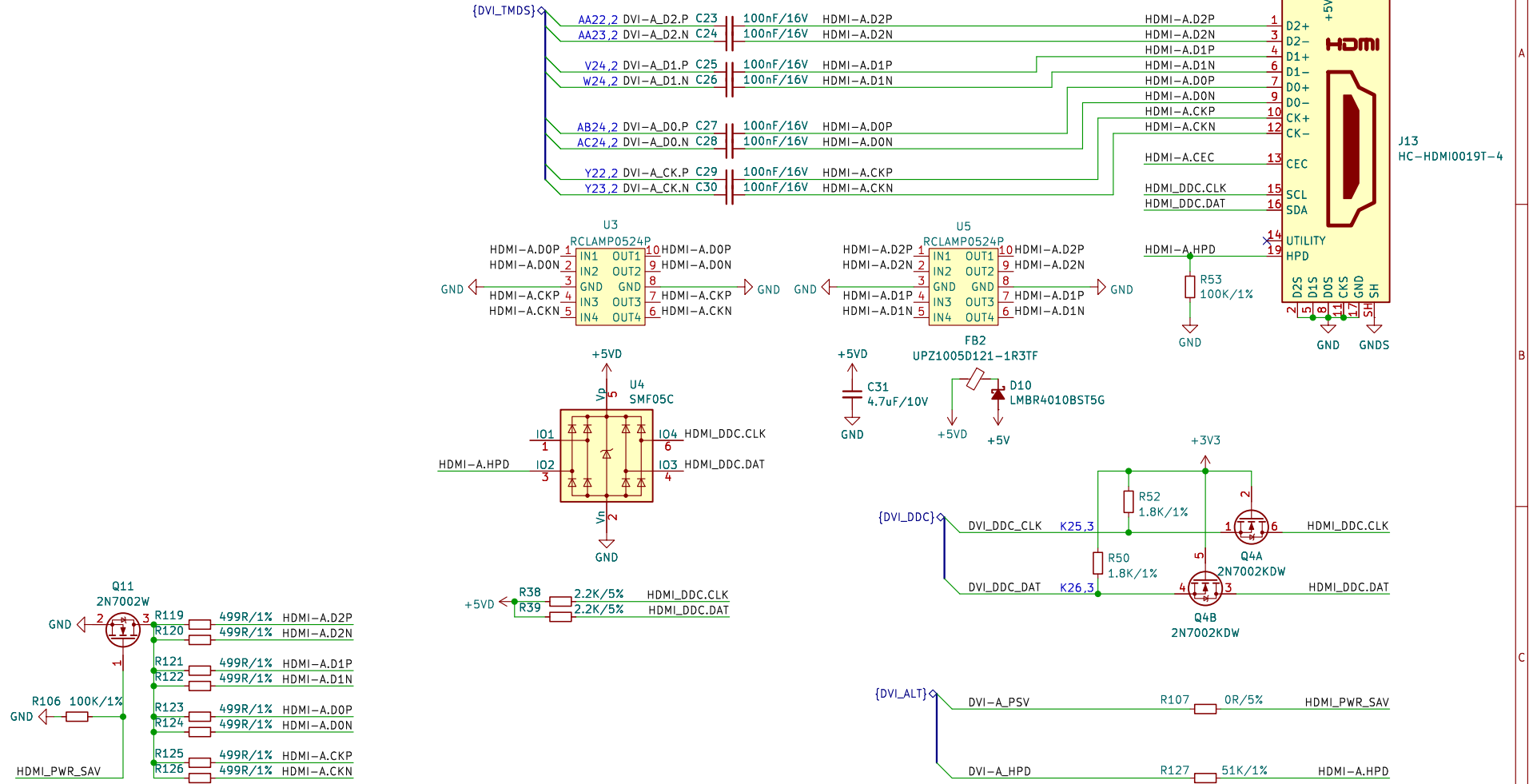
KiCad E.D.A. 9.0.1

Rev: 1.22

Id: 8/17



# DVI OUTPUT



A Retro Ready Console Powered by FPGA

**Shenzhen Sipeed Technology Co.,Ltd.**

Sheet: /DVI\_TMDS/

File: TMDS\_TX.kicad\_sch

**Title: Tang\_60K\_Console\_V32001A**

Size: A4

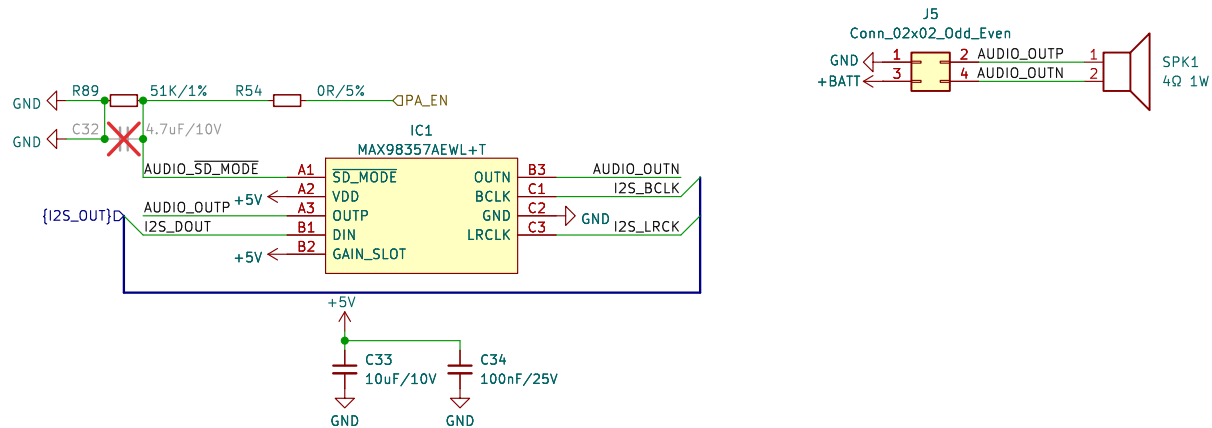
Date: 2024-12-31

KiCad E.D.A. 9.0.1

**Rev: 1.22**

Id: 9/17

## DAC + Audio PA



A Retro Ready Console Powered by FPGA

**Shenzhen Sipeed Technology Co.,Ltd.**

Sheet: /AUDIO+PA/

File: AUDIO.kicad\_sch

**Title: Tang\_60K\_Console\_V32001A**

Size: A4

Date: 2024-12-31

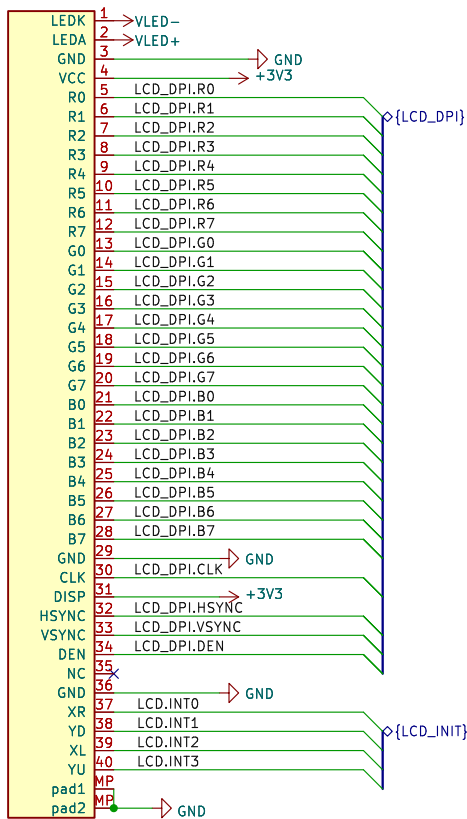
Rev: 1.22

KiCad E.D.A. 9.0.1

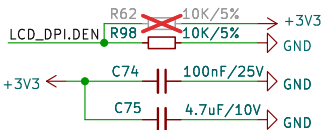
Id: 10/17

# RGB LCD Display

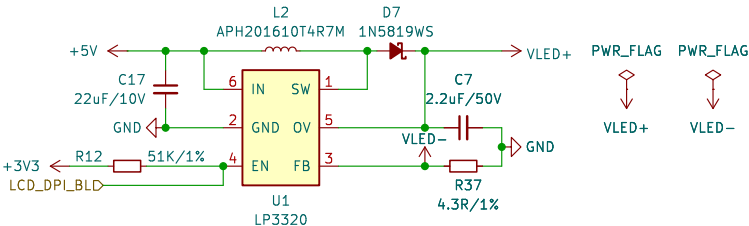
## RGB LCD Display Connector



J3  
FPC-40P-0.5MM-H2.0MM-rearlock



## BACKLIGHT Driver



Feedback Voltage = 250mV  
BackLight Current =  $250\text{mV}/4.3\text{R} \approx 58.1\text{mA}$   
7inch LCD QTM070TFT40-001 : IF=20\*3=60mA

LCD.INT1 R13 5.1K/1% → +3V3  
LCD.INT0 R36 5.1K/1% → +3V3



A Retro Ready Console Powered by FPGA

**Shenzhen Sipeed Technology Co.,Ltd.**

Sheet: /LCD\_DPI\_24BIT/

File: LCD\_DPI.kicad\_sch

**Title: Tang\_60K\_Console\_V32001A**

Size: A4

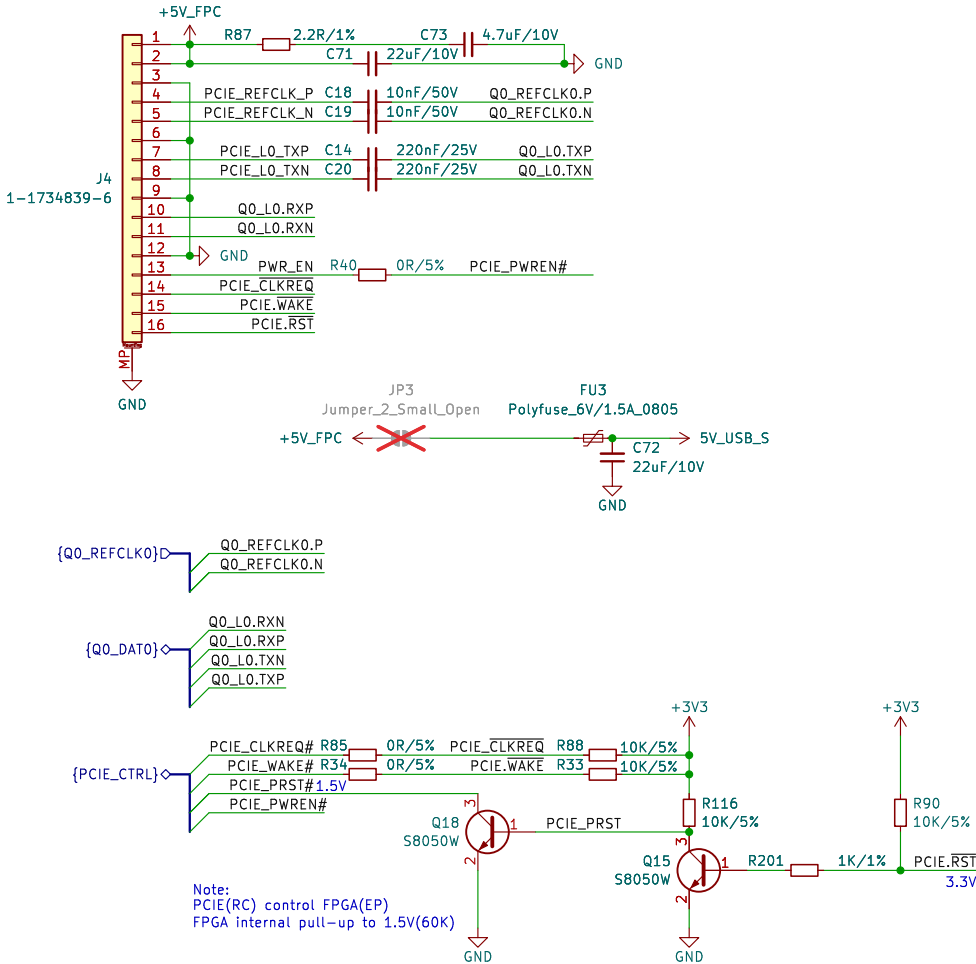
Date: 2024-12-31

Rev: 1.22

KiCad E.D.A. 9.0.1

Id: 11/17

RPI HAT+ PCIE



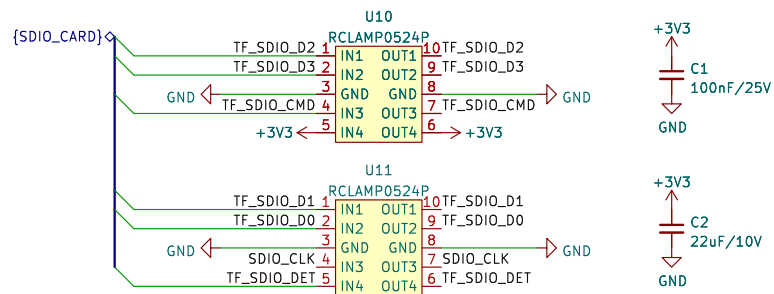
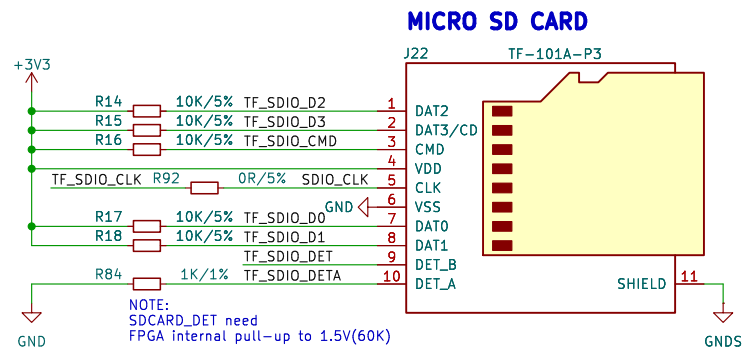
A Retro Ready Console Powered by FPGA  
Shenzhen Sipeed Technology Co.,Ltd.

Sheet: /PCIE\_GT/  
File: SERDES.kicad\_sch

Title: Tang\_60K\_Console\_V32001A

Size: A4 Date: 2024-12-31  
KiCad E.D.A. 9.0.1

Rev: 1.22  
Id: 12/17



A Retro Ready Console Powered by FPGA

**Shenzhen Sippeed Technology Co.,Ltd.**

Sheet: /SDIO\_CARD/

File: SDCARD.kicad\_sch

**Title: Tang\_60K\_Console\_V32001A**

Size: A4

Date: 2024-12-31

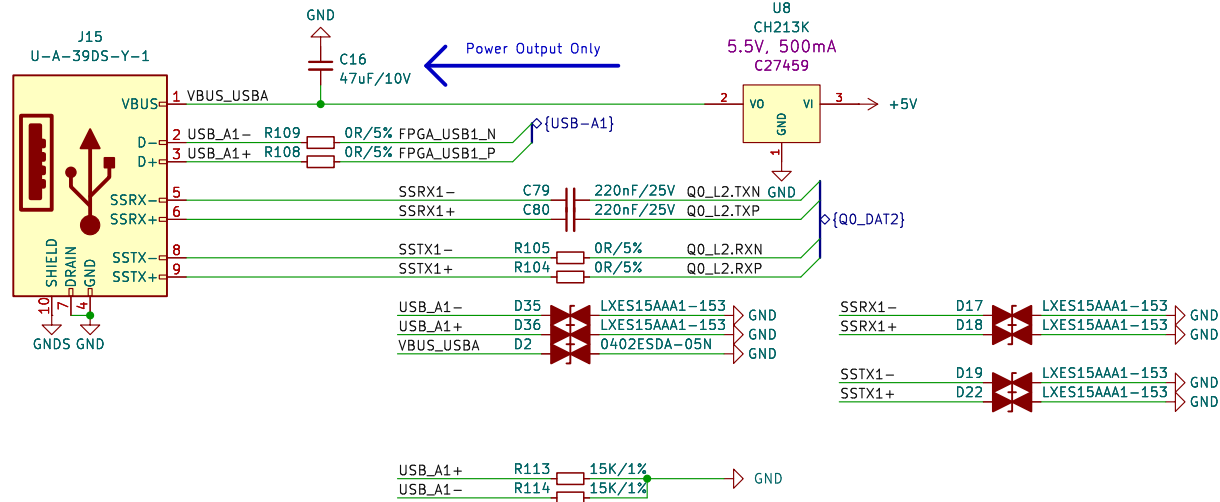
Rev: 1.22

KiCad E.D.A. 9.0.1

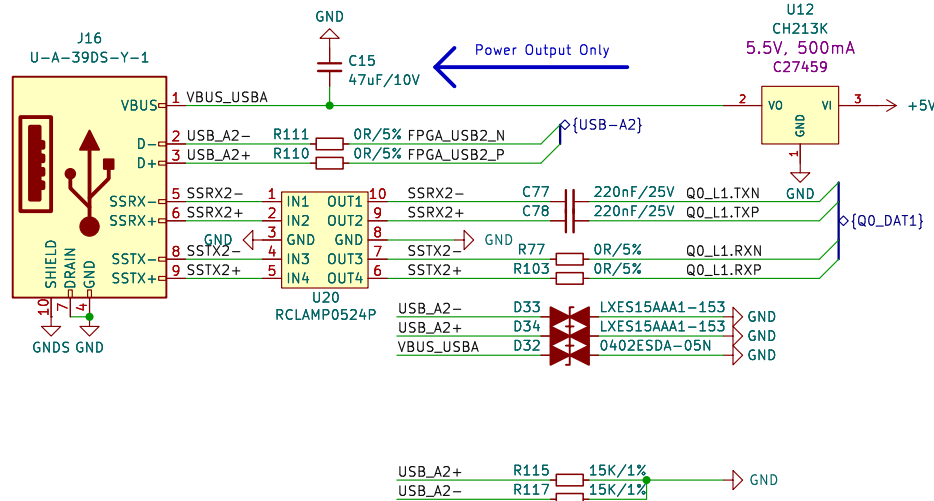
Id: 14/17

# USB\_Ax2

## USB\_A1 Full-Speed Host + Super-Speed Device



## USB\_A2 Full-Speed Host + Super-Speed Device



A Retro Ready Console Powered by FPGA

**Shenzhen Sipeed Technology Co.,Ltd.**

Sheet: /USB3-A\_GT/

File: USB3-GT.kicad\_sch

**Title: Tang\_60K\_Console\_V32001A**

Size: A4 Date: 2024-12-31

KiCad E.D.A. 9.0.1

**Rev: 1.22**

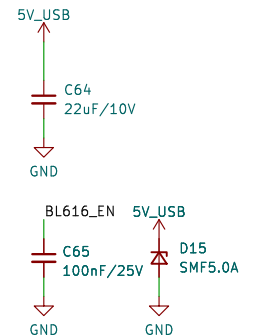
Id: 14/17

```
{DBG_TWI} ◇
```

Power Input only



JP7  



**Shenzhen Sipeeed Technology Co.,Ltd.**

Sheet: /USB\_JTAG/

File: USB\_JTAG.kicad\_sch

**Title:** Tang\_60K\_Console\_V32001A

|          |                  |
|----------|------------------|
| Size: A4 | Date: 2024-12-31 |
|----------|------------------|

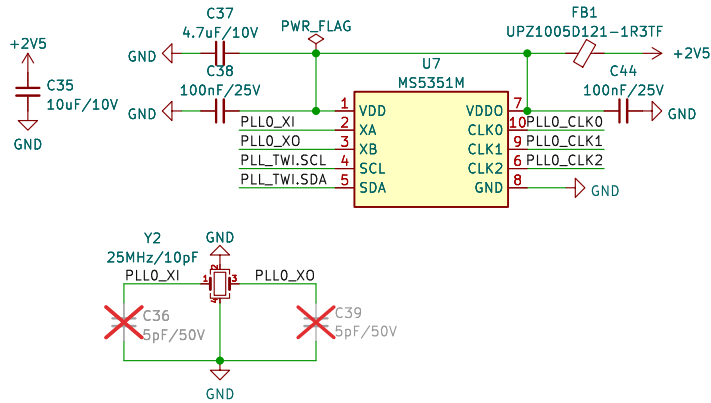
KiCad E.D.A. 9.0.1

Rev: 1.22

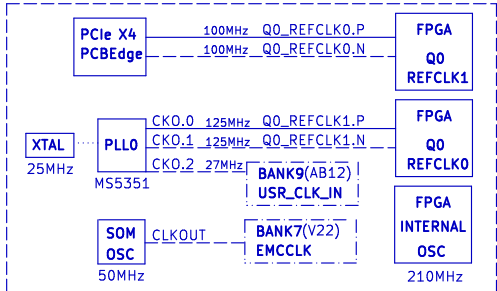
Id: 16/17

## CLOCK

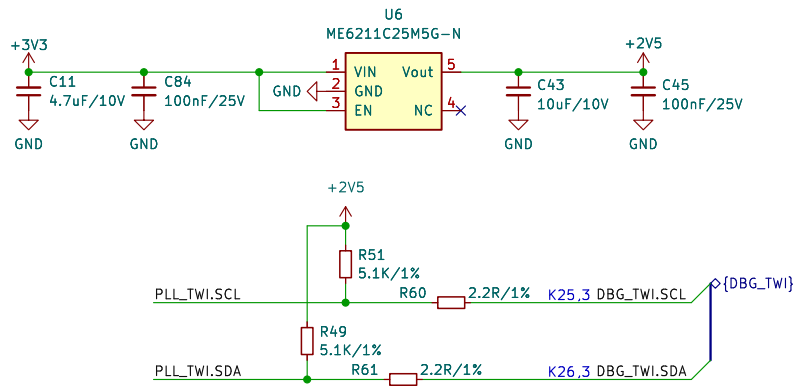
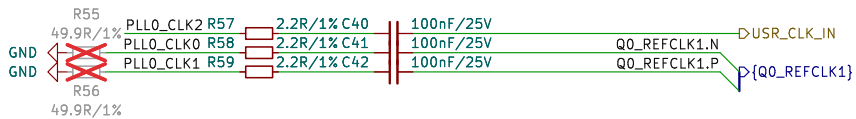
## EXT. PL



## COLCK TREE



FPGA internal clock generator OSC.  
The user can change the value of  
the option "Frequency Divider" to  
generate a different output clock  
frequency, which can be one of  
64 values: 1/2, 1/3, 1/4, 1/6, 1/8, 1/10,  
..... or 1/126 of the oscillator frequency.



## A Retro Ready Console Powered by FPGA

Shenzhen Sipeeed Technology Co.,Ltd.

Sheet: /SYS\_CLOCK/

File: CLOCK.kicad\_sch

**Title:** Tang\_60K\_Console\_V32001A

|          |                  |
|----------|------------------|
| Size: A4 | Date: 2024-12-31 |
|----------|------------------|

KiCad E.D.A. 9.0.1

Rev: 1.22

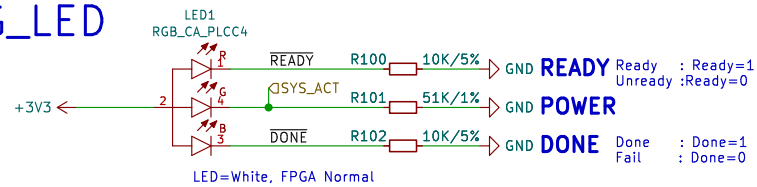
Id: 18/17



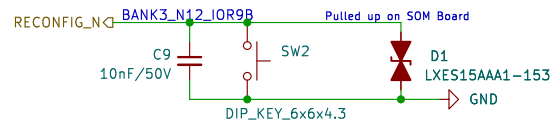
## Inverter Circuit



## CFG\_LED



## RECFG\_KEY



A Retro Ready Console Powered by FPGA

**Shenzhen Sipeed Technology Co.,Ltd.**

Sheet: /FPAG\_CONFIG/

File: FPGA\_CFG.kicad\_sch

**Title: Tang\_60K\_Console\_V32001A**

Size: A4

Date: 2024-12-31

Rev: 1.22

KiCad E.D.A. 9.0.1

Id: 19/17